

Bitwise differential trail through **QALQAN** found by SAT

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1 QALQAN

Maximal differential probability: 2^{-22}

0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	0	3	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	0	3	
KeyAddStart																													
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	0	3	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	0	3	
QalqaRound																													
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0
SBoxLayer																													
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	0	0
L																													
0	0	0	0	0	0	0	8	0	0	0	0	0	0	8	0	0	0	0	0	0	8	0	0	0	0	8	0	8	0
0	0	0	0	0	0	0	8	0	0	0	0	0	0	8	0	0	0	0	0	0	8	0	0	0	0	8	0	8	0
KeyAddFin																													
0	0	0	0	0	0	0	8	0	0	0	0	0	0	8	0	0	0	0	0	0	8	0	0	0	0	8	0	8	0
KeyAddFin																													
0	0	0	0	0	0	0	8	0	0	0	0	0	0	8	0	0	0	0	0	0	8	0	0	0	0	8	0	8	0

2 QalqanRound

Maximal differential probability: $2^{-15.0}$

0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	0	3
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	0	3
S Box Layer																												
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	2	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	2	0
L																												
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0
Add128																												
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0

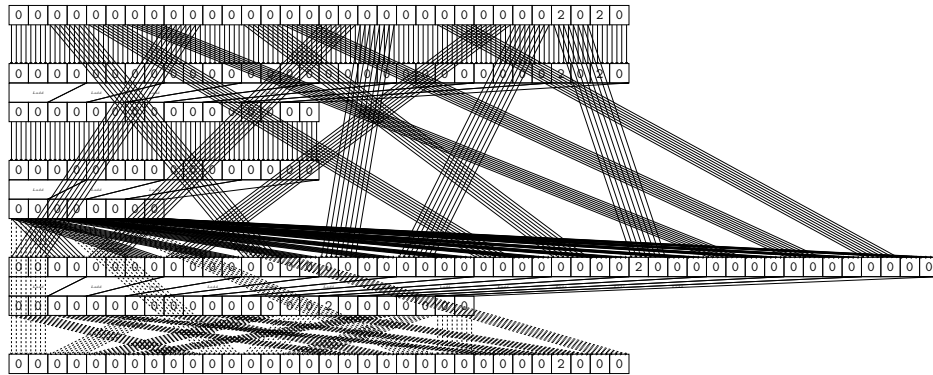
3 SBoxLayer

Maximal differential probability: $2^{-12.0}$

0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	0	3	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	0	3	
<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>		
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	2	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	2	0

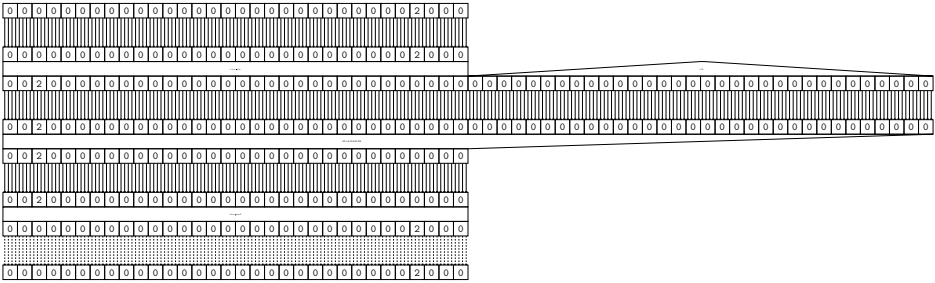
4 L

Maximal differential probability: $2^{-2.0}$



5 Add128

Maximal differential probability: $2^{-1.0}$



6 SBoxLayer

Maximal differential probability: $2^{-7.0}$ [illegible]

7 L

Maximal differential probability: $2^{-0.0}$

